Image 209-timeline-layout-b-workbench.png

What it shows: Layout B (Workbench): windows and sources in a side panel, figures and a taller chart on the right.
How to reach it: Prototype controls: layout pill, choose B.

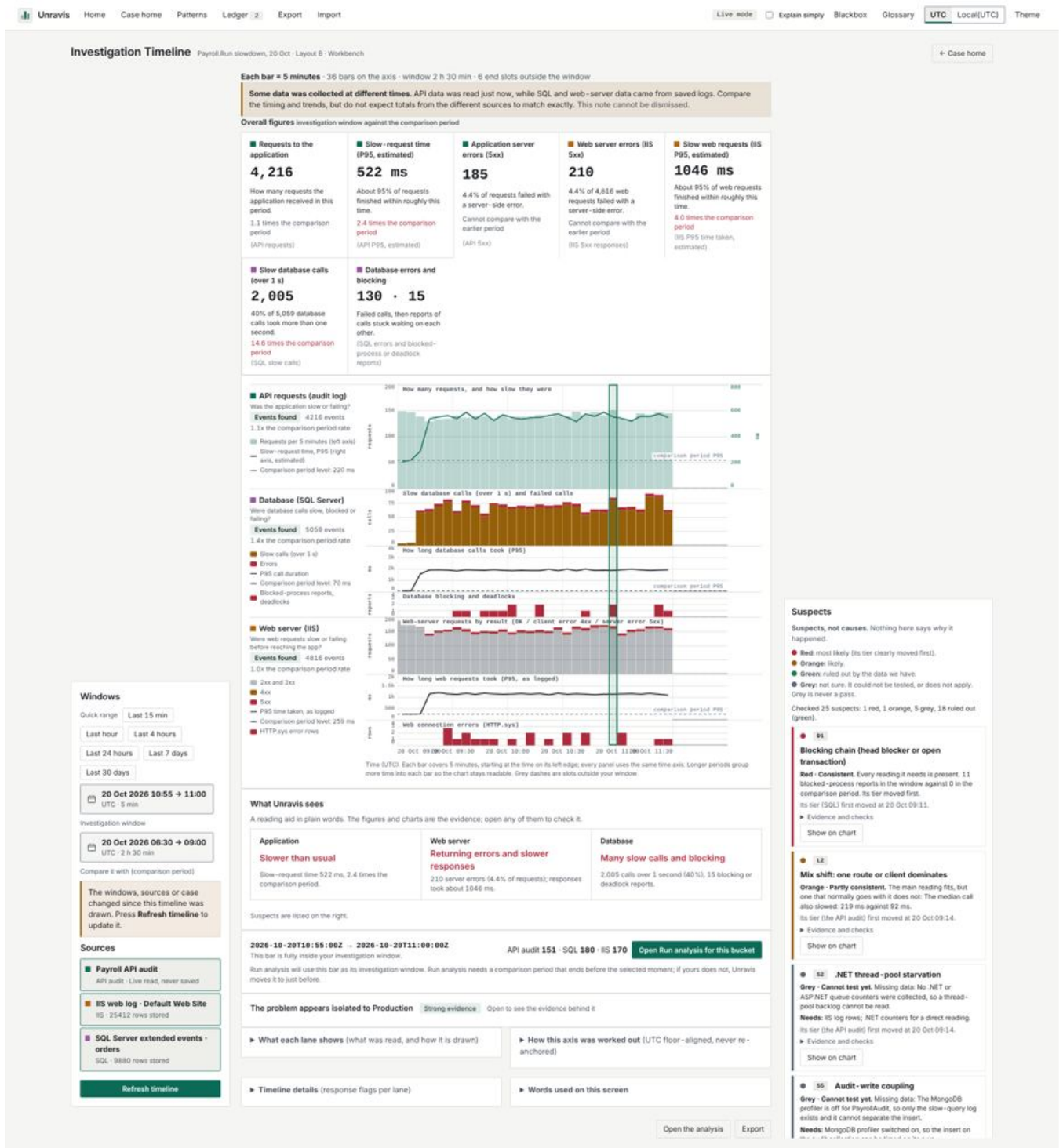


Image 210-timeline-layout-c-lane-ledger.png

What it shows: Layout C (Lane ledger): one row per lane with trust stickers, event count and a sparkline; open a lane for its panels.
How to reach it: Prototype controls: layout pill, choose C.

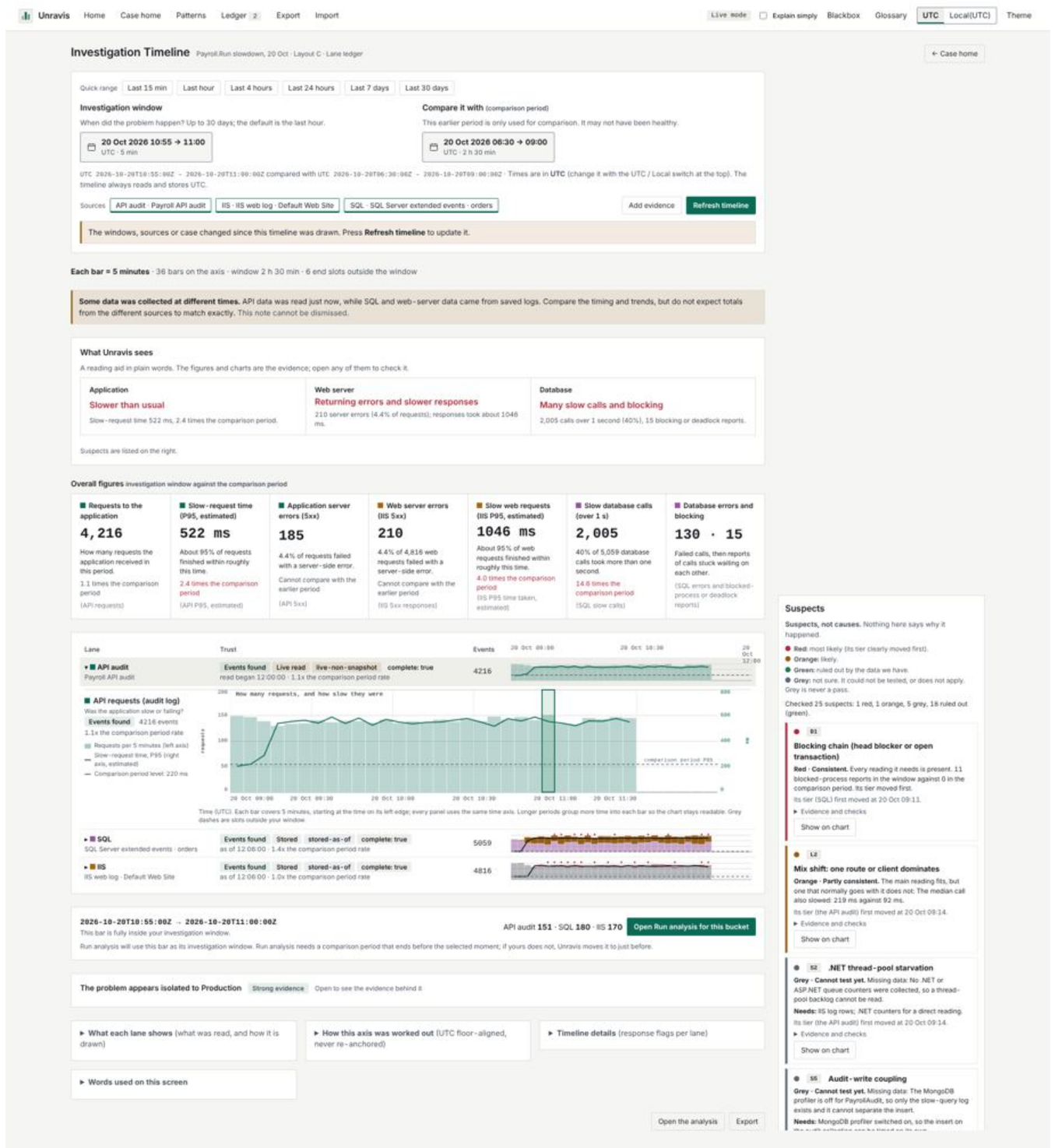


Image 211-suspects-rail-top-five.png

What it shows: The Suspects rail: legend, summary line, and the top five suspects (red first, then orange, then grey).
How to reach it: Timeline, right side.

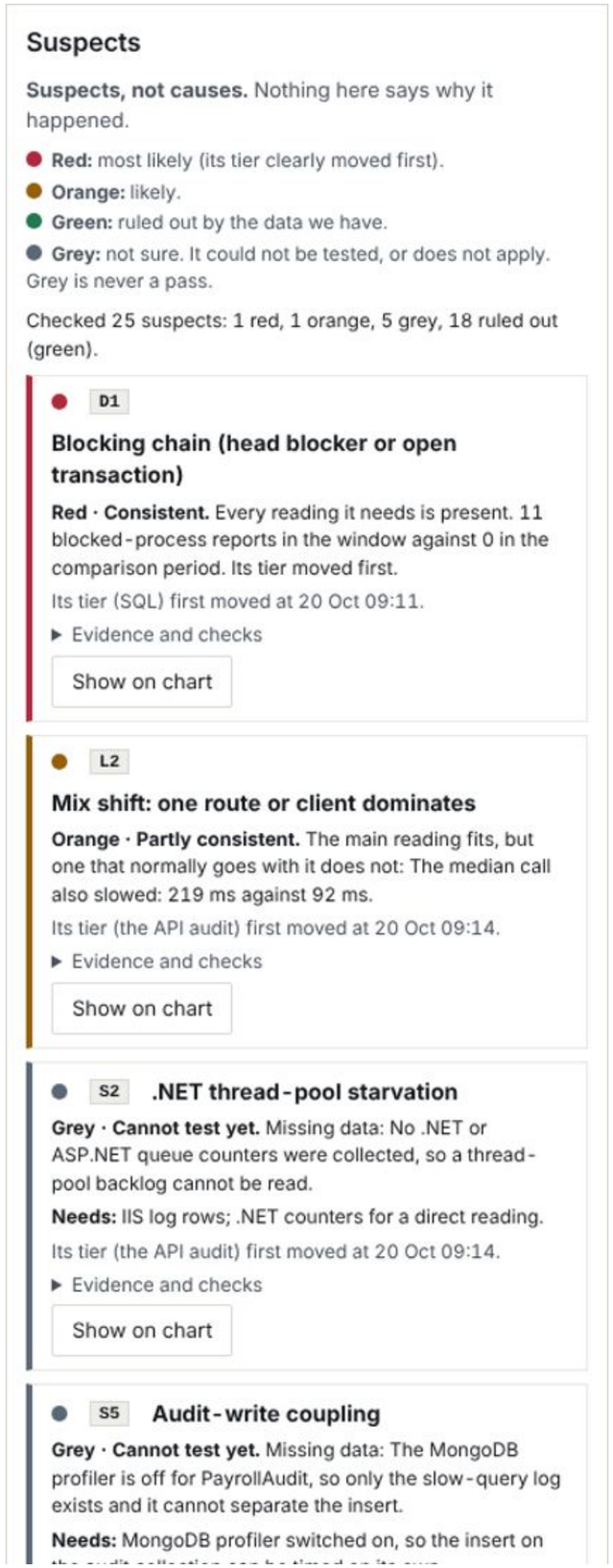


Image 212-suspects-rail-see-all.png

What it shows: See all expands in place: every suspect, red then orange then grey, green ones folded as Ruled out by the data.
How to reach it: Press See all.

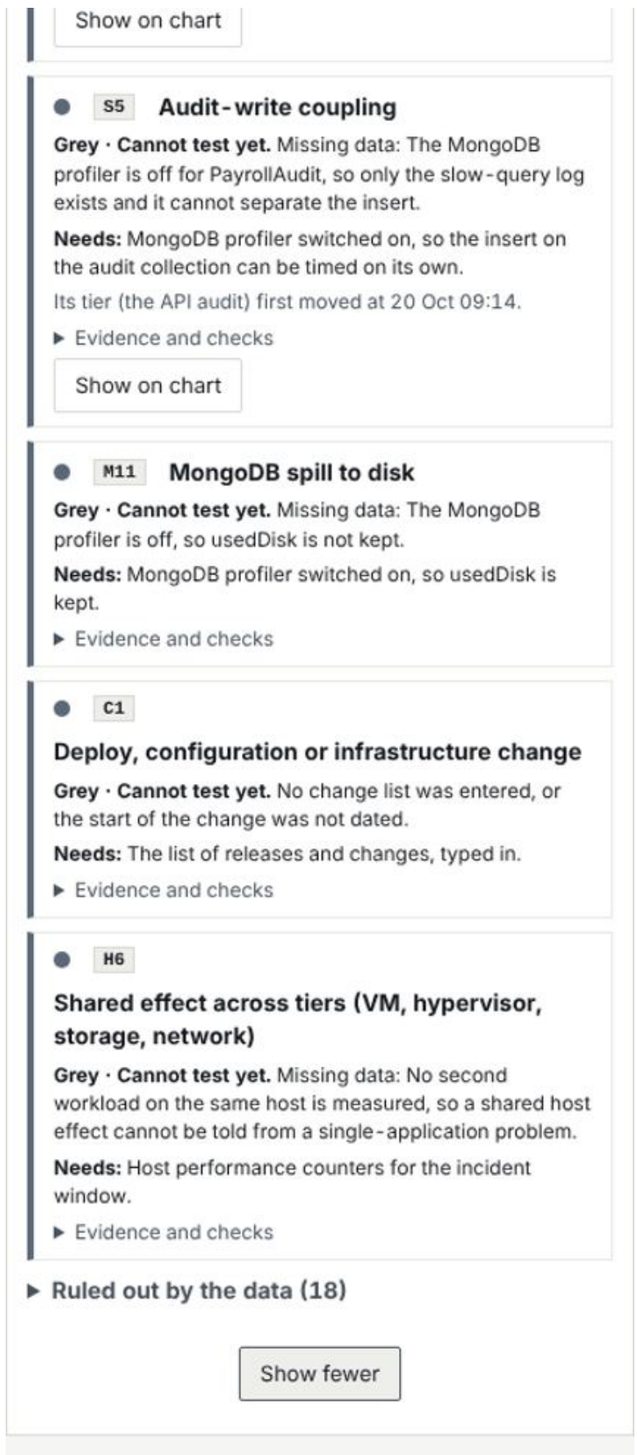
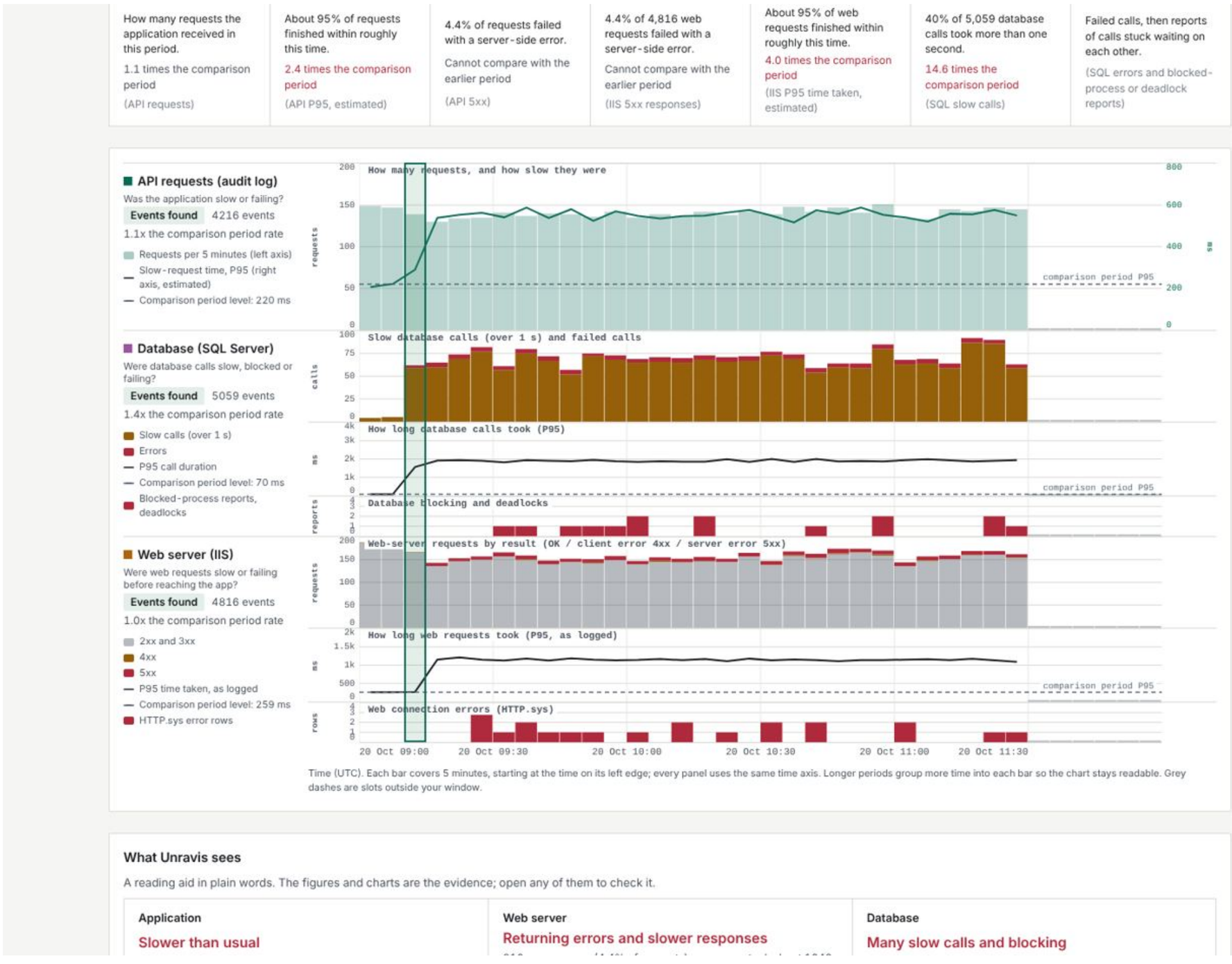


Image 213-suspects-show-on-chart.png

What it shows: Show on chart selects the bucket where that suspect first moved.
How to reach it: Press Show on chart on the red suspect.



What Unravis sees

A reading aid in plain words. The figures and charts are the evidence; open any of them to check it.

Application	Web server	Database
Slower than usual	Returning errors and slower responses	Many slow calls and blocking

Suspects

Suspects, not causes. Nothing here says why it happened.

- **Red:** most likely (its tier clearly moved first).
- **Orange:** likely.
- **Green:** ruled out by the data we have.
- **Grey:** not sure. It could not be tested, or does not apply. Grey is never a pass.

Checked 25 suspects: 1 red, 1 orange, 5 grey, 18 ruled out (green).

D1

Blocking chain (head blocker or open transaction)

Red · Consistent. Every reading it needs is present. 11 blocked-process reports in the window against 0 in the comparison period. Its tier moved first.

Its tier (SQL) first moved at 20 Oct 09:11.

▸ Evidence and checks

Show on chart

L2

Mix shift: one route or client dominates

Orange · Partly consistent. The main reading fits, but one that normally goes with it does not: The median call also slowed: 219 ms against 92 ms.

Its tier (the API audit) first moved at 20 Oct 09:14.

▸ Evidence and checks

Show on chart

S2 **.NET thread-pool starvation**

Grey · Cannot test yet. Missing data: No .NET or ASP.NET queue counters were collected, so a thread-pool backlog cannot be read.

Needs: IIS log rows; .NET counters for a direct reading.

Its tier (the API audit) first moved at 20 Oct 09:14.

▸ Evidence and checks

Show on chart

S5 **Audit-write coupling**

Grey · Cannot test yet. Missing data: The MongoDB profiler is off for PayrollAudit, so only the slow-query log exists and it cannot separate the insert.

Needs: MongoDB profiler switched on, so the insert on

Image 214-suspects-ruled-out-folded.png

What it shows: Green suspects are folded away under Ruled out by the data.
How to reach it: Open the Ruled out by the data fold at the bottom of the list.

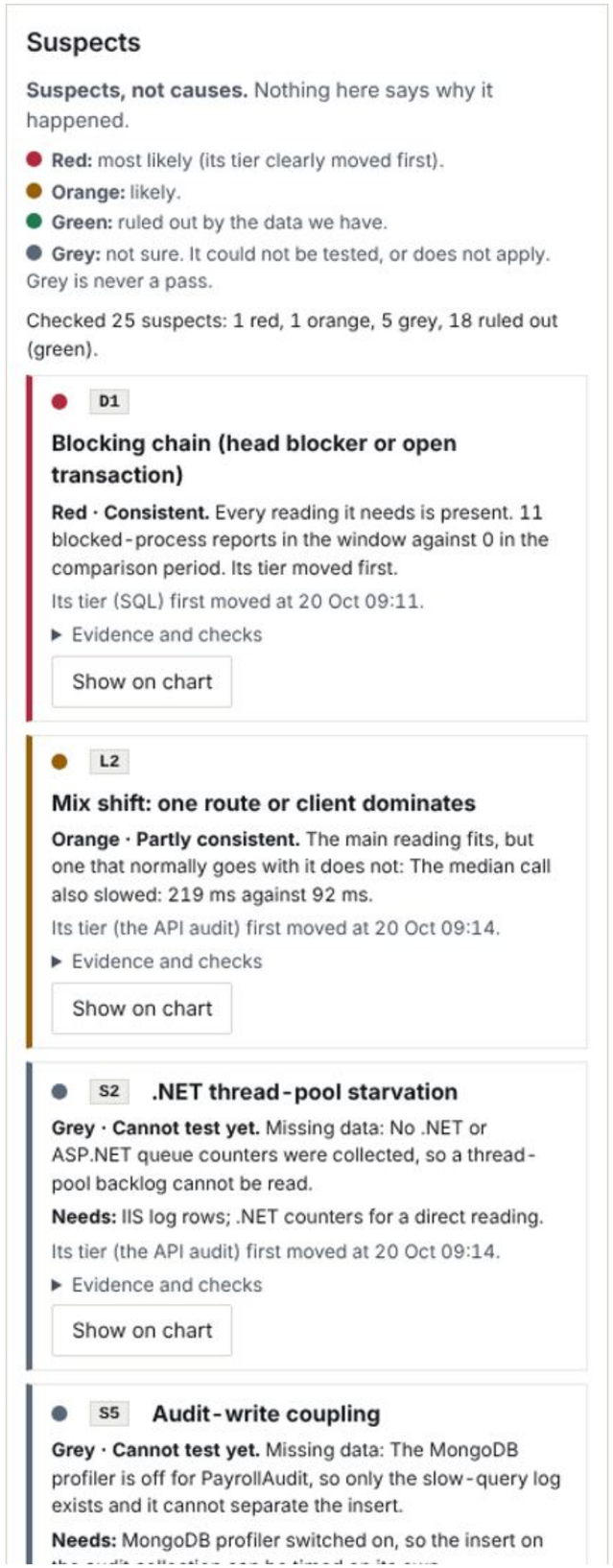


Image 215-suspects-show-fewer.png

What it shows: Show fewer returns to the top five.
How to reach it: Press Show fewer.

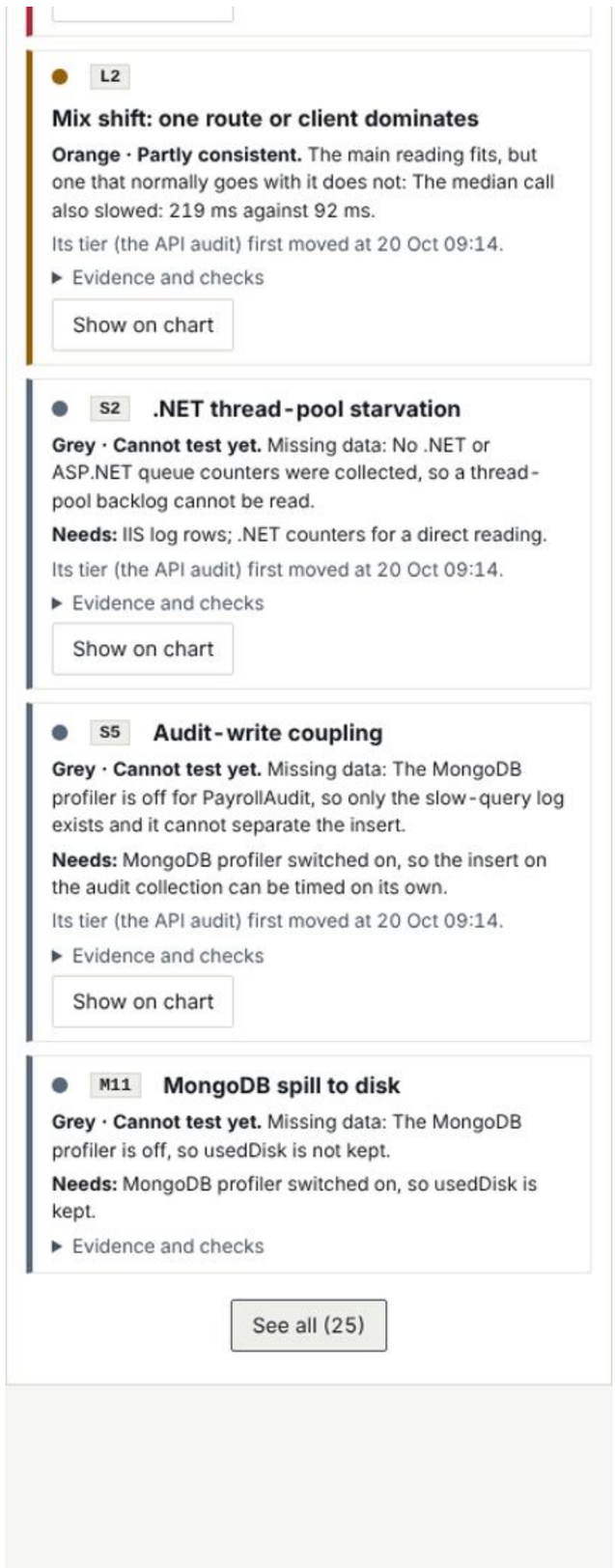


Image 216-suspects-card-open-evidence.png

What it shows: One suspect opened: evidence rows (signal, Must or Expect, yes, no or unknown, why), the cheapest check and what would overturn it.
How to reach it: Open the evidence on a suspect card.

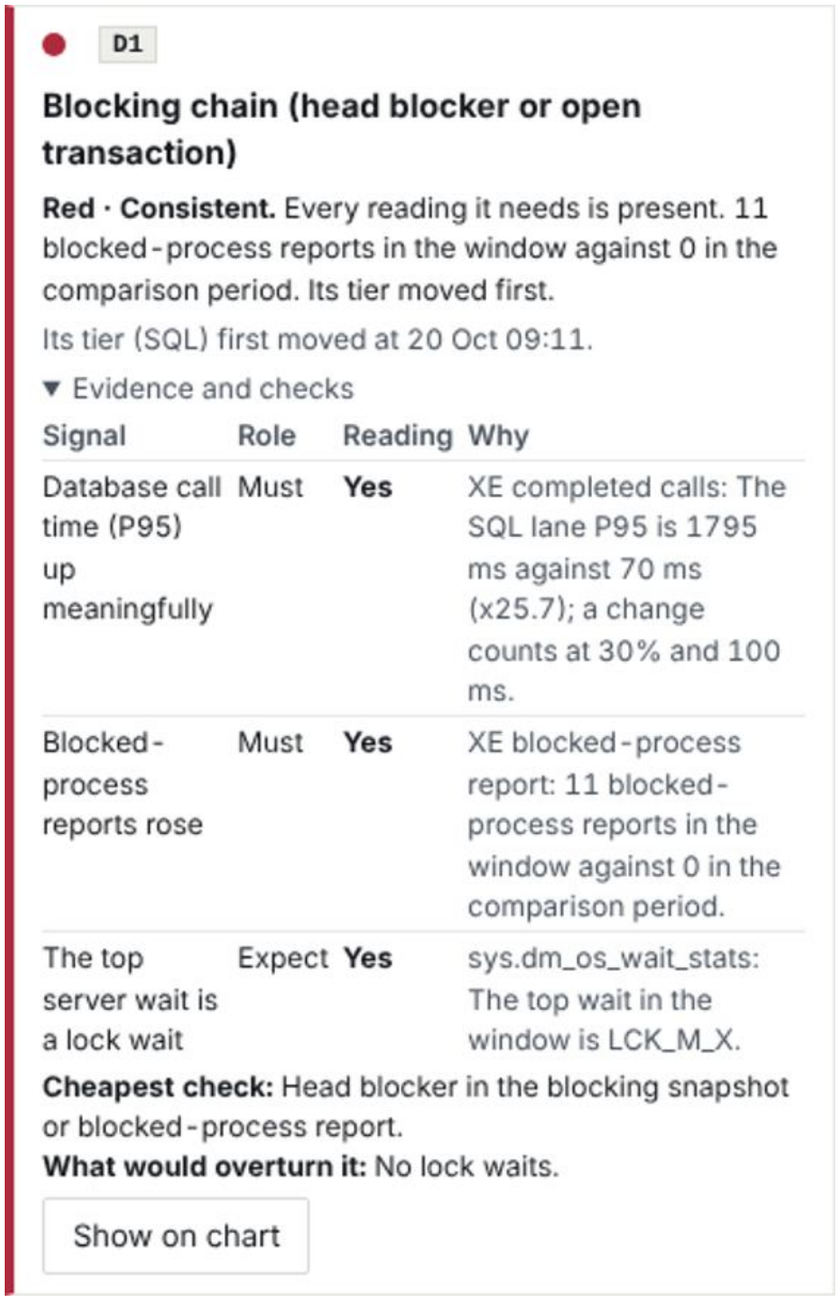


Image 217-evidence-card-folded.png

What it shows: The Where does the evidence point card, folded: title and strength chip only.
How to reach it: Scroll below the chart.

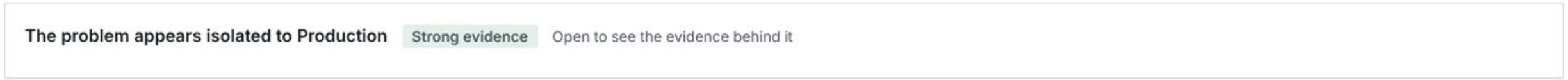


Image 218-evidence-card-open-one-environment.png

What it shows: Opened: The problem appears isolated to Production, with the server block and the by-database table.
How to reach it: Open the folded card.

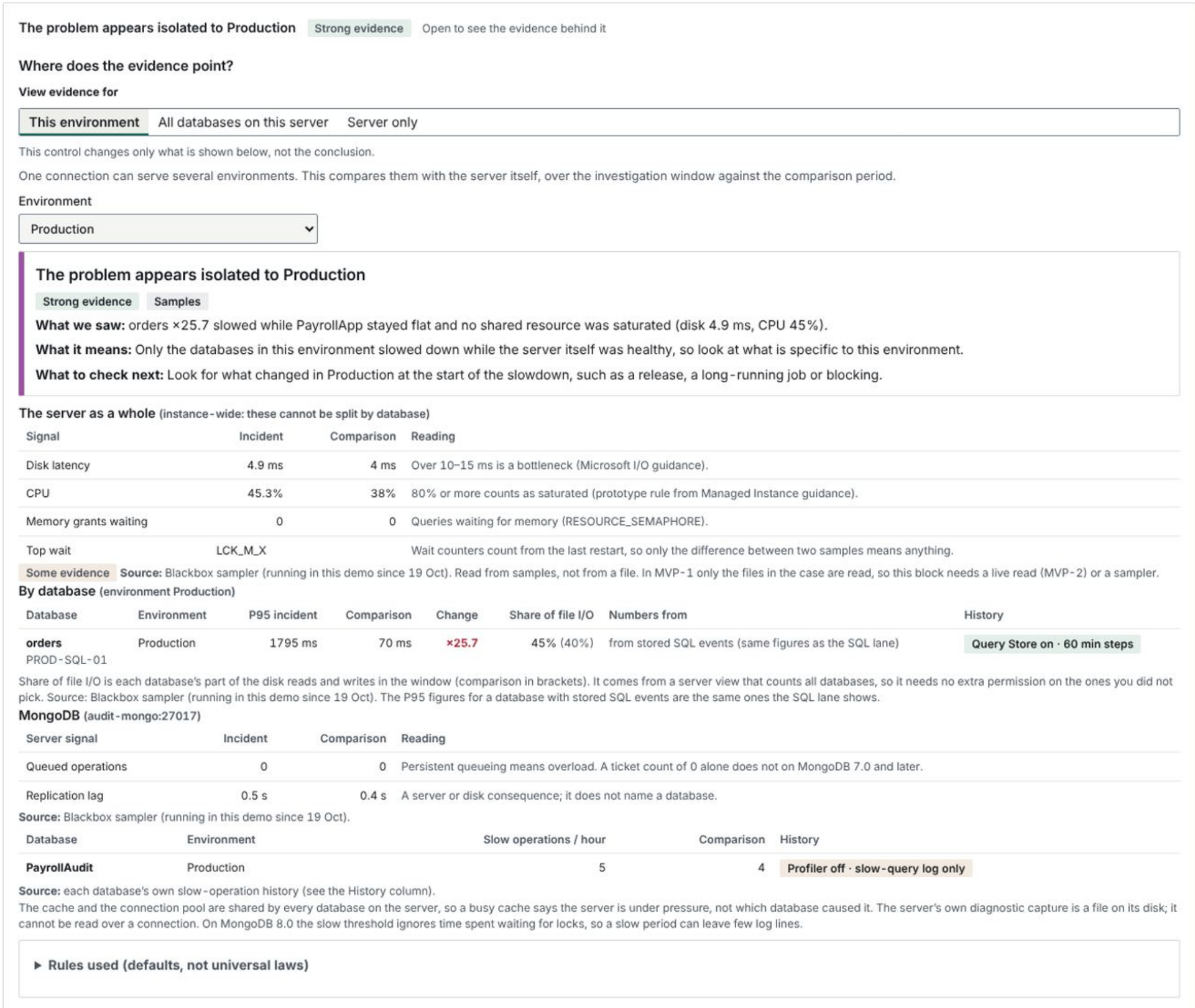
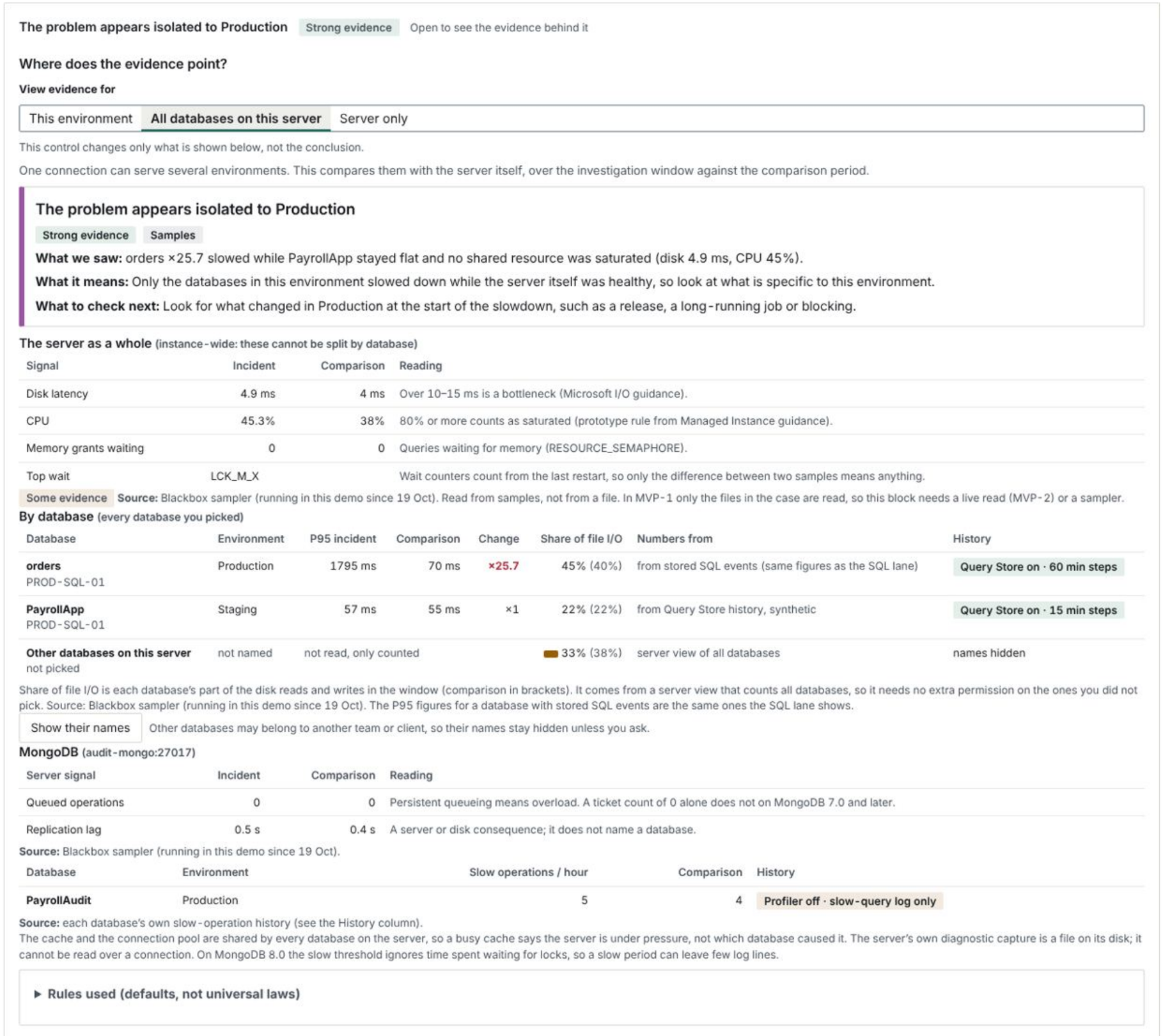


Image 219-evidence-scope-all-databases.png

What it shows: Scope switch: All databases on this server, with an unnamed Other databases row.
How to reach it: Press All databases on this server.



The problem appears isolated to Production Strong evidence Open to see the evidence behind it

Where does the evidence point?

View evidence for

This environment | **All databases on this server** | Server only

This control changes only what is shown below, not the conclusion.

One connection can serve several environments. This compares them with the server itself, over the investigation window against the comparison period.

The problem appears isolated to Production

Strong evidence Samples

What we saw: orders ×25.7 slowed while PayrollApp stayed flat and no shared resource was saturated (disk 4.9 ms, CPU 45%).

What it means: Only the databases in this environment slowed down while the server itself was healthy, so look at what is specific to this environment.

What to check next: Look for what changed in Production at the start of the slowdown, such as a release, a long-running job or blocking.

The server as a whole (instance-wide: these cannot be split by database)

Signal	Incident	Comparison	Reading
Disk latency	4.9 ms	4 ms	Over 10–15 ms is a bottleneck (Microsoft I/O guidance).
CPU	45.3%	38%	80% or more counts as saturated (prototype rule from Managed Instance guidance).
Memory grants waiting	0	0	Queries waiting for memory (RESOURCE_SEMAPHORE).
Top wait	LCK_M_X		Wait counters count from the last restart, so only the difference between two samples means anything.

Some evidence **Source:** Blackbox sampler (running in this demo since 19 Oct). Read from samples, not from a file. In MVP-1 only the files in the case are read, so this block needs a live read (MVP-2) or a sampler.

By database (every database you picked)

Database	Environment	P95 incident	Comparison	Change	Share of file I/O	Numbers from	History
orders PROD-SQL-01	Production	1795 ms	70 ms	×25.7	45% (40%)	from stored SQL events (same figures as the SQL lane)	Query Store on · 60 min steps
PayrollApp PROD-SQL-01	Staging	57 ms	55 ms	×1	22% (22%)	from Query Store history, synthetic	Query Store on · 15 min steps
Other databases on this server not picked	not named	not read, only counted			33% (38%)	server view of all databases	names hidden

Share of file I/O is each database's part of the disk reads and writes in the window (comparison in brackets). It comes from a server view that counts all databases, so it needs no extra permission on the ones you did not pick. Source: Blackbox sampler (running in this demo since 19 Oct). The P95 figures for a database with stored SQL events are the same ones the SQL lane shows.

Show their names — Other databases may belong to another team or client, so their names stay hidden unless you ask.

MongoDB (audit-mongo:27017)

Server signal	Incident	Comparison	Reading
Queued operations	0	0	Persistent queueing means overload. A ticket count of 0 alone does not on MongoDB 7.0 and later.
Replication lag	0.5 s	0.4 s	A server or disk consequence; it does not name a database.

Source: Blackbox sampler (running in this demo since 19 Oct).

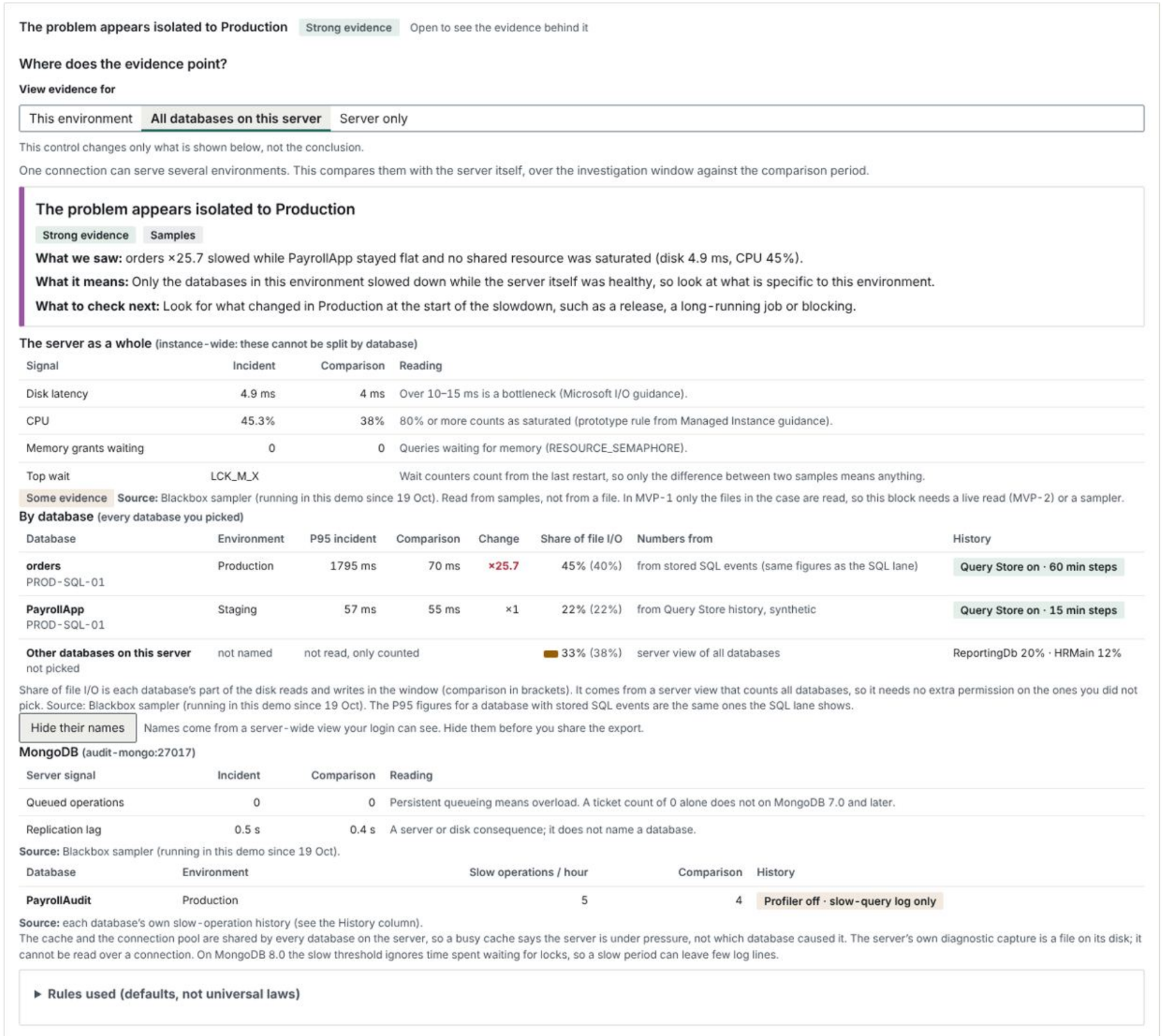
Database	Environment	Slow operations / hour	Comparison	History
PayrollAudit	Production	5	4	Profiler off · slow-query log only

Source: each database's own slow-operation history (see the History column).
The cache and the connection pool are shared by every database on the server, so a busy cache says the server is under pressure, not which database caused it. The server's own diagnostic capture is a file on its disk; it cannot be read over a connection. On MongoDB 8.0 the slow threshold ignores time spent waiting for locks, so a slow period can leave few log lines.

▸ **Rules used (defaults, not universal laws)**

Image 220-evidence-scope-show-names.png

What it shows: Show their names reveals the unpicked databases, with a warning to hide them before sharing an export.
How to reach it: Press Show their names.



The problem appears isolated to Production Strong evidence Open to see the evidence behind it

Where does the evidence point?

View evidence for

This environment | **All databases on this server** | Server only

This control changes only what is shown below, not the conclusion.

One connection can serve several environments. This compares them with the server itself, over the investigation window against the comparison period.

The problem appears isolated to Production

Strong evidence Samples

What we saw: orders ×25.7 slowed while PayrollApp stayed flat and no shared resource was saturated (disk 4.9 ms, CPU 45%).

What it means: Only the databases in this environment slowed down while the server itself was healthy, so look at what is specific to this environment.

What to check next: Look for what changed in Production at the start of the slowdown, such as a release, a long-running job or blocking.

The server as a whole (instance-wide: these cannot be split by database)

Signal	Incident	Comparison	Reading
Disk latency	4.9 ms	4 ms	Over 10–15 ms is a bottleneck (Microsoft I/O guidance).
CPU	45.3%	38%	80% or more counts as saturated (prototype rule from Managed Instance guidance).
Memory grants waiting	0	0	Queries waiting for memory (RESOURCE_SEMAPHORE).
Top wait	LCK_M_X		Wait counters count from the last restart, so only the difference between two samples means anything.

Some evidence **Source:** Blackbox sampler (running in this demo since 19 Oct). Read from samples, not from a file. In MVP-1 only the files in the case are read, so this block needs a live read (MVP-2) or a sampler.

By database (every database you picked)

Database	Environment	P95 incident	Comparison	Change	Share of file I/O	Numbers from	History
orders PROD-SQL-01	Production	1795 ms	70 ms	×25.7	45% (40%)	from stored SQL events (same figures as the SQL lane)	Query Store on · 60 min steps
PayrollApp PROD-SQL-01	Staging	57 ms	55 ms	×1	22% (22%)	from Query Store history, synthetic	Query Store on · 15 min steps
Other databases on this server not picked	not named	not read, only counted			33% (38%)	server view of all databases	ReportingDb 20% · HRMain 12%

Share of file I/O is each database's part of the disk reads and writes in the window (comparison in brackets). It comes from a server view that counts all databases, so it needs no extra permission on the ones you did not pick. Source: Blackbox sampler (running in this demo since 19 Oct). The P95 figures for a database with stored SQL events are the same ones the SQL lane shows.

Hide their names Names come from a server-wide view your login can see. Hide them before you share the export.

MongoDB (audit-mongo:27017)

Server signal	Incident	Comparison	Reading
Queued operations	0	0	Persistent queueing means overload. A ticket count of 0 alone does not on MongoDB 7.0 and later.
Replication lag	0.5 s	0.4 s	A server or disk consequence; it does not name a database.

Source: Blackbox sampler (running in this demo since 19 Oct).

Database	Environment	Slow operations / hour	Comparison	History
PayrollAudit	Production	5	4	Profiler off · slow-query log only

Source: each database's own slow-operation history (see the History column).
The cache and the connection pool are shared by every database on the server, so a busy cache says the server is under pressure, not which database caused it. The server's own diagnostic capture is a file on its disk; it cannot be read over a connection. On MongoDB 8.0 the slow threshold ignores time spent waiting for locks, so a slow period can leave few log lines.

▸ **Rules used (defaults, not universal laws)**

Image 221-evidence-scope-server-only.png

What it shows: Scope switch: Server only.
How to reach it: Press Server only.

The problem appears isolated to Production Strong evidence Open to see the evidence behind it

Where does the evidence point?

View evidence for

This environment | All databases on this server | **Server only**

This control changes only what is shown below, not the conclusion.

One connection can serve several environments. This compares them with the server itself, over the investigation window against the comparison period.

The problem appears isolated to Production

Strong evidence Samples

What we saw: orders ×25.7 slowed while PayrollApp stayed flat and no shared resource was saturated (disk 4.9 ms, CPU 45%).

What it means: Only the databases in this environment slowed down while the server itself was healthy, so look at what is specific to this environment.

What to check next: Look for what changed in Production at the start of the slowdown, such as a release, a long-running job or blocking.

The server as a whole (instance-wide: these cannot be split by database)

Signal	Incident	Comparison	Reading
Disk latency	4.9 ms	4 ms	Over 10–15 ms is a bottleneck (Microsoft I/O guidance).
CPU	45.3%	38%	80% or more counts as saturated (prototype rule from Managed Instance guidance).
Memory grants waiting	0	0	Queries waiting for memory (RESOURCE_SEMAPHORE).
Top wait	LCK_M_X		Wait counters count from the last restart, so only the difference between two samples means anything.

Some evidence **Source:** Blackbox sampler (running in this demo since 19 Oct). Read from samples, not from a file. In MVP-1 only the files in the case are read, so this block needs a live read (MVP-2) or a sampler.

MongoDB (audit-mongo:27017)

Server signal	Incident	Comparison	Reading
Queued operations	0	0	Persistent queueing means overload. A ticket count of 0 alone does not on MongoDB 7.0 and later.
Replication lag	0.5 s	0.4 s	A server or disk consequence; it does not name a database.

Source: Blackbox sampler (running in this demo since 19 Oct).
The cache and the connection pool are shared by every database on the server, so a busy cache says the server is under pressure, not which database caused it. The server's own diagnostic capture is a file on its disk; it cannot be read over a connection. On MongoDB 8.0 the slow threshold ignores time spent waiting for locks, so a slow period can leave few log lines.

▶ **Rules used (defaults, not universal laws)**

Image 222-evidence-verdict-whole-server.png

What it shows: The whole database server appears affected (every picked database slowed, disk saturated).
How to reach it: Scenario a-server (Investigation Timeline: where does the evidence point).

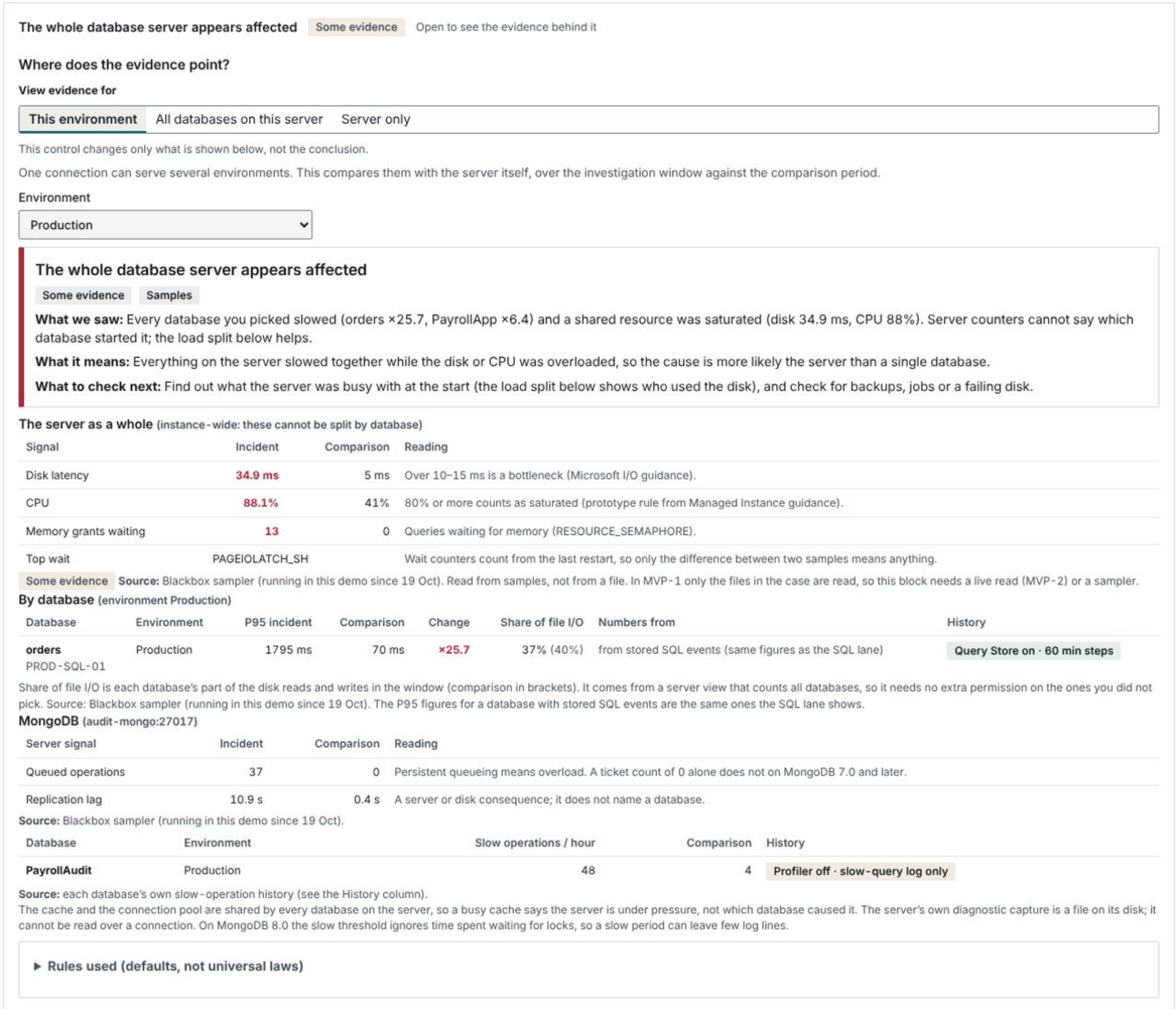


Image 223-evidence-verdict-noisy-neighbour.png

What it shows: Another database may be affecting yours (unpicked databases take most of the disk).
How to reach it: Scenario a-noisy (Investigation Timeline: where does the evidence point).

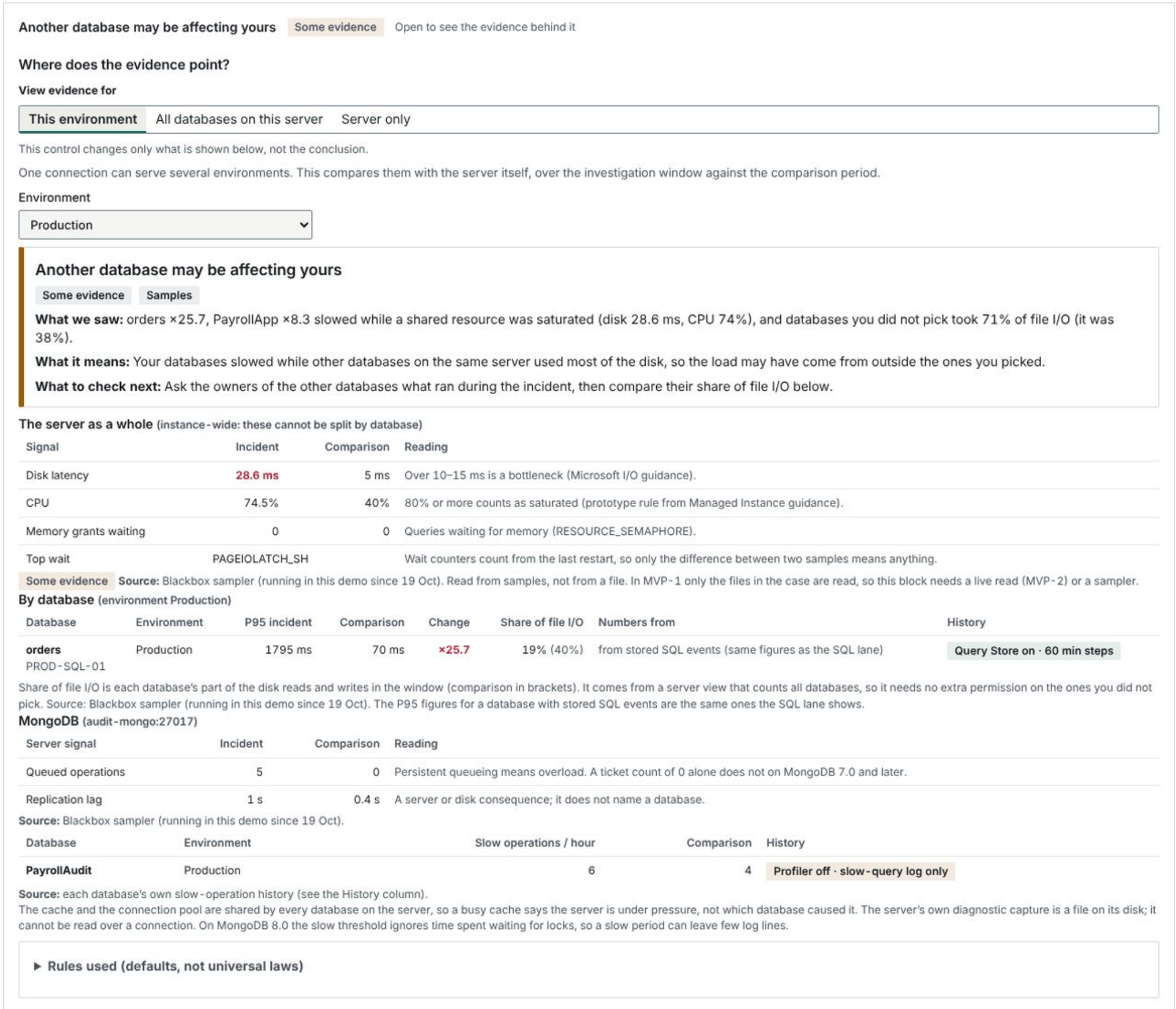


Image 224-evidence-verdict-not-enough-evidence.png

What it shows: There is not enough evidence to narrow it down (Query Store off, no samples).
How to reach it: Scenario a-blind (Investigation Timeline: where does the evidence point).

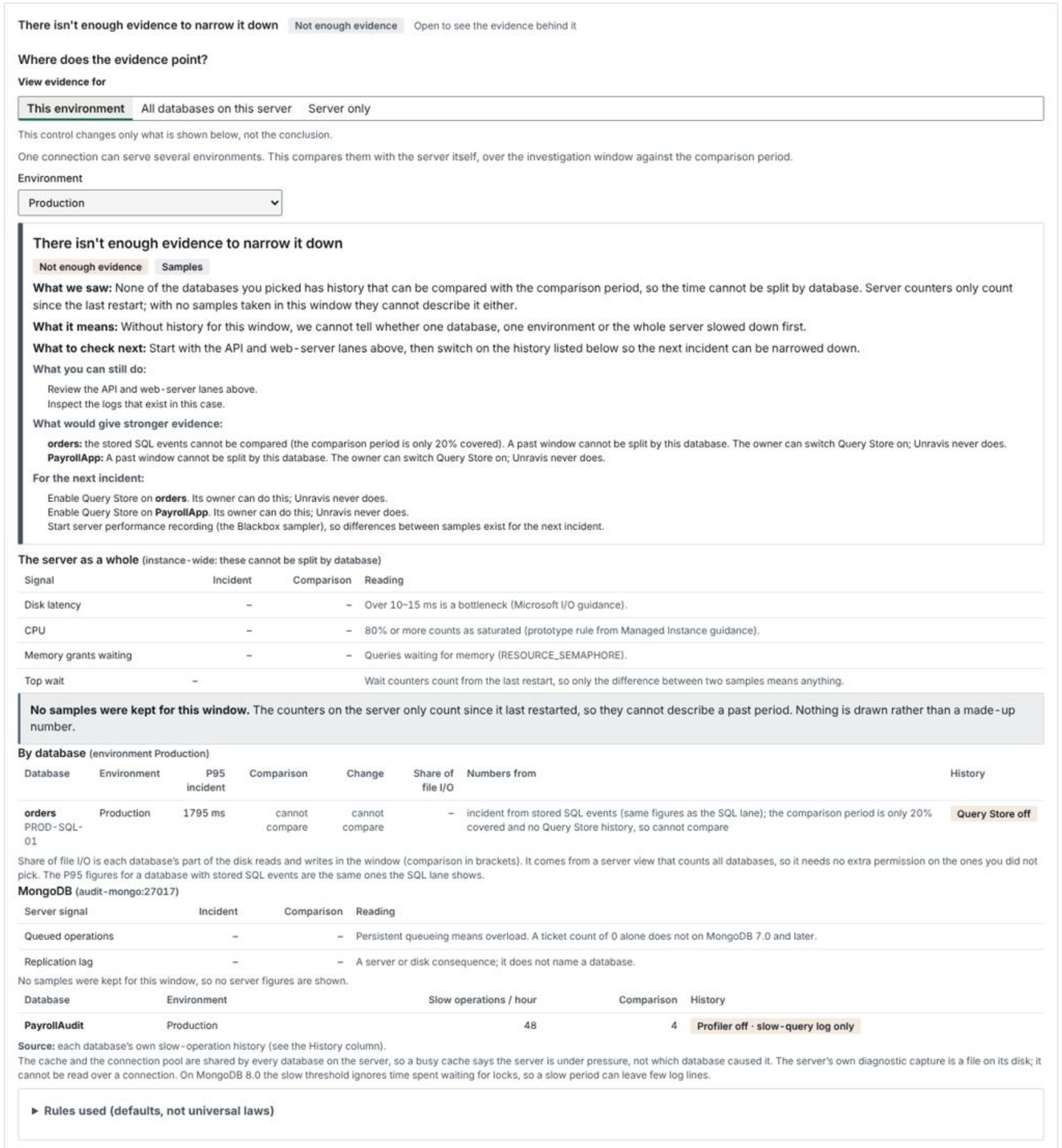


Image 225-evidence-verdict-quiet-window.png

What it shows: No clear performance problem in this period.
How to reach it: Scenario a-quiet (Investigation Timeline: where does the evidence point).

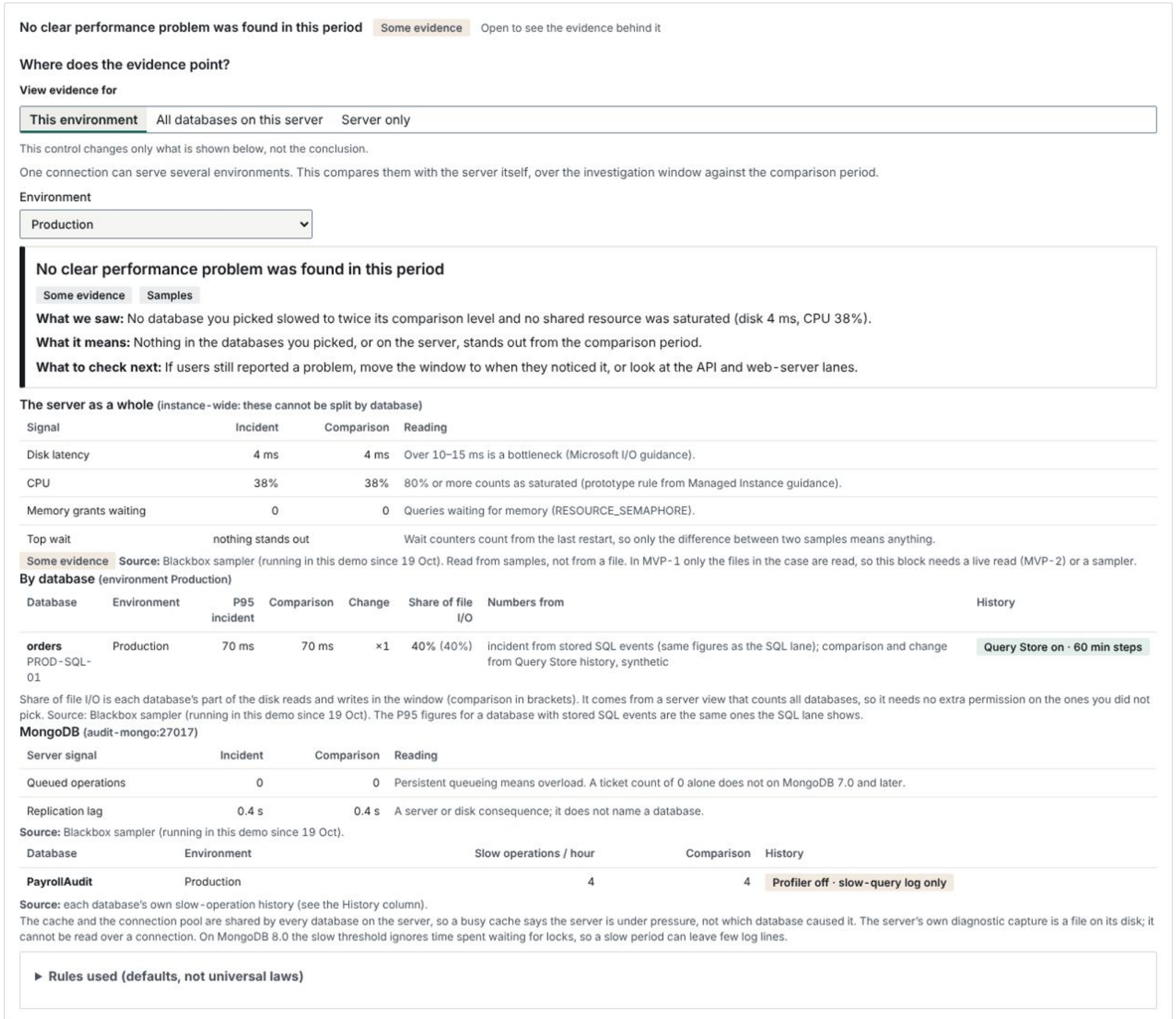


Image 226-timeline-no-history-full.png

What it shows: No history (nothing kept): most suspects are grey with a Needs line, nothing is red.
How to reach it: Scenario a-blind: the whole screen.

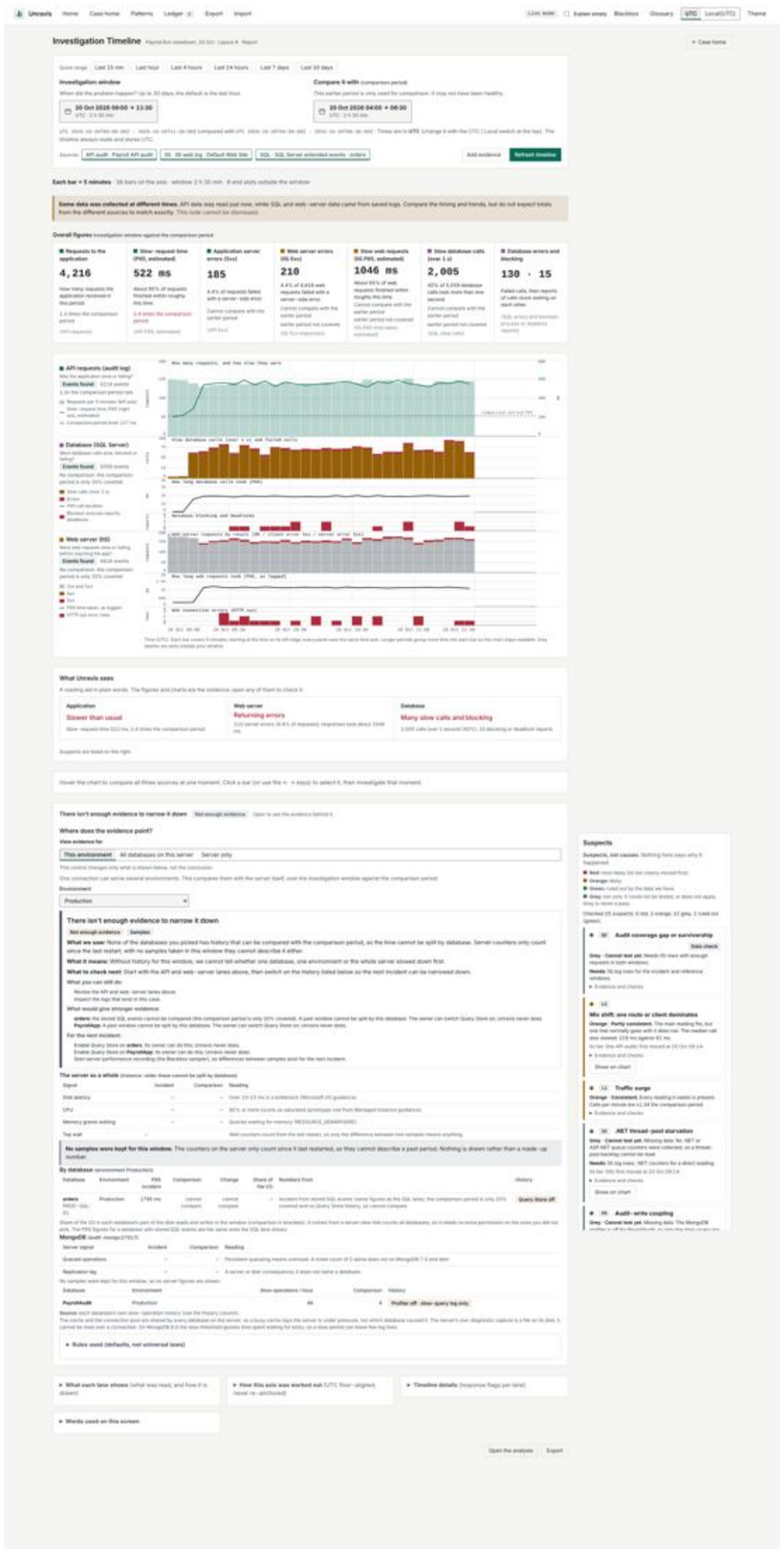


Image 227-suspects-rail-no-history.png

What it shows: Suspects rail in the no-history situation.
How to reach it: Scenario a-blind.

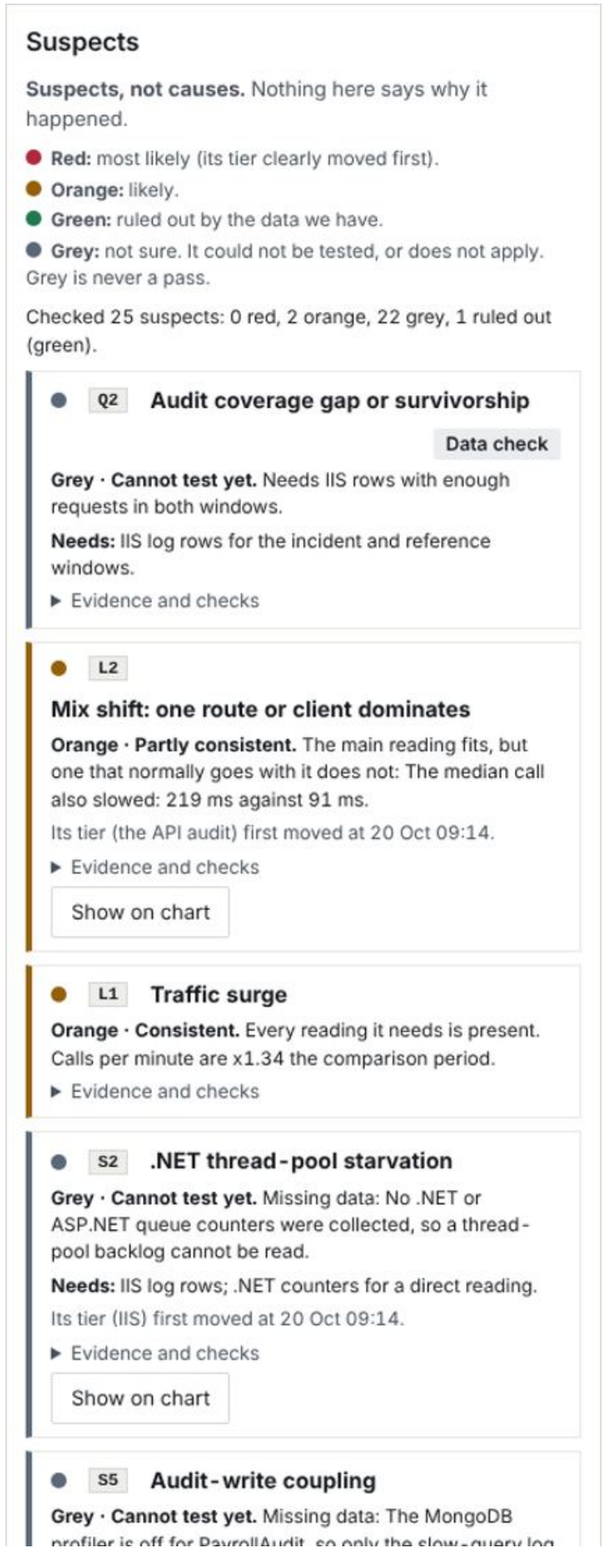


Image 228-timeline-one-environment-full.png

What it shows: One environment: D1 blocking chain is the only red suspect.
How to reach it: Scenario a-env.

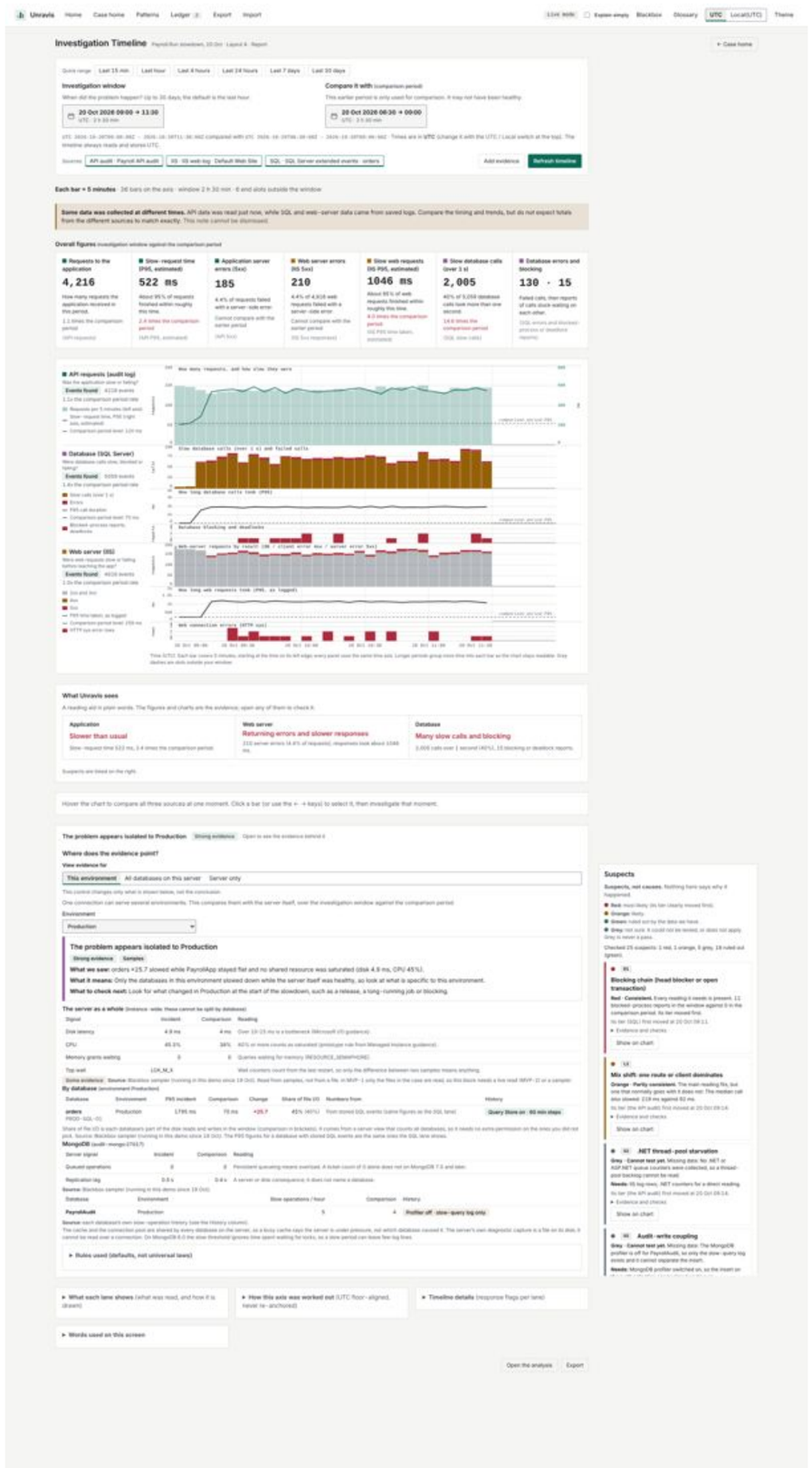


Image 229-timeline-seven-days-gaps.png

What it shows: Last 7 days: 2-hour bars, SQL and IIS say there are no logs for most of the window (hatched, with exact range), the audit outage is a thin gap.
How to reach it: Press Last 7 days.

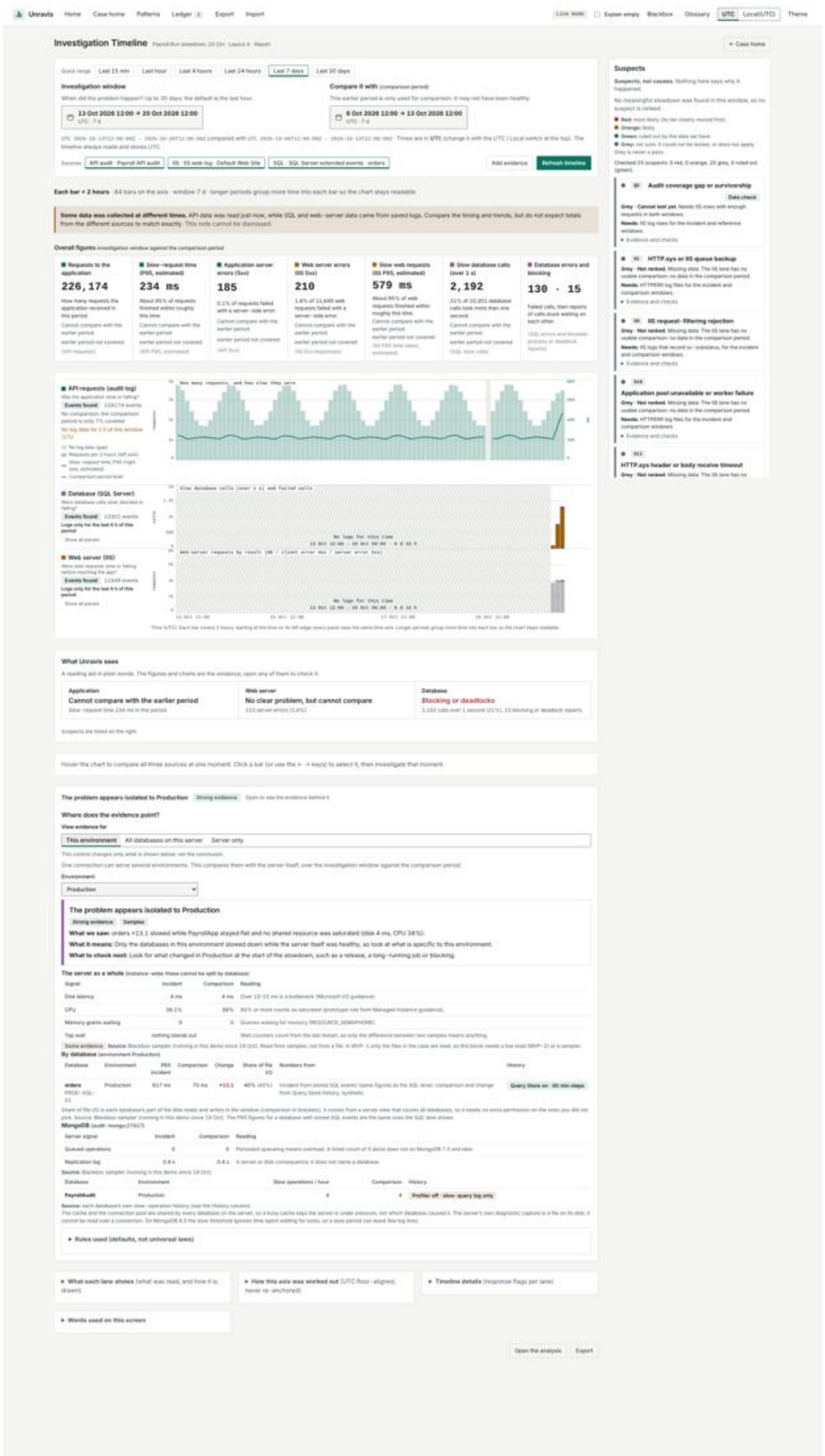
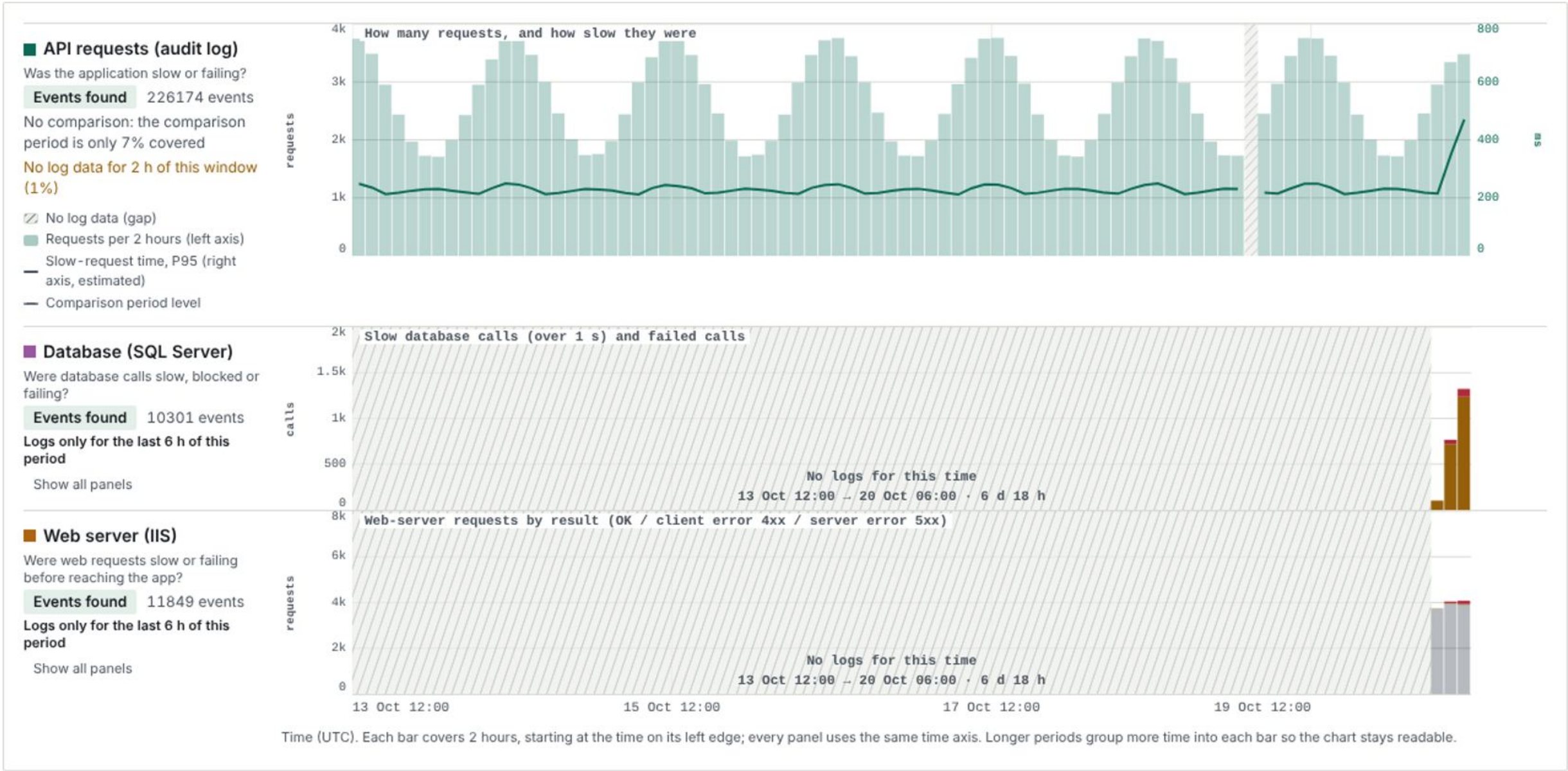


Image 230-timeline-seven-days-board.png

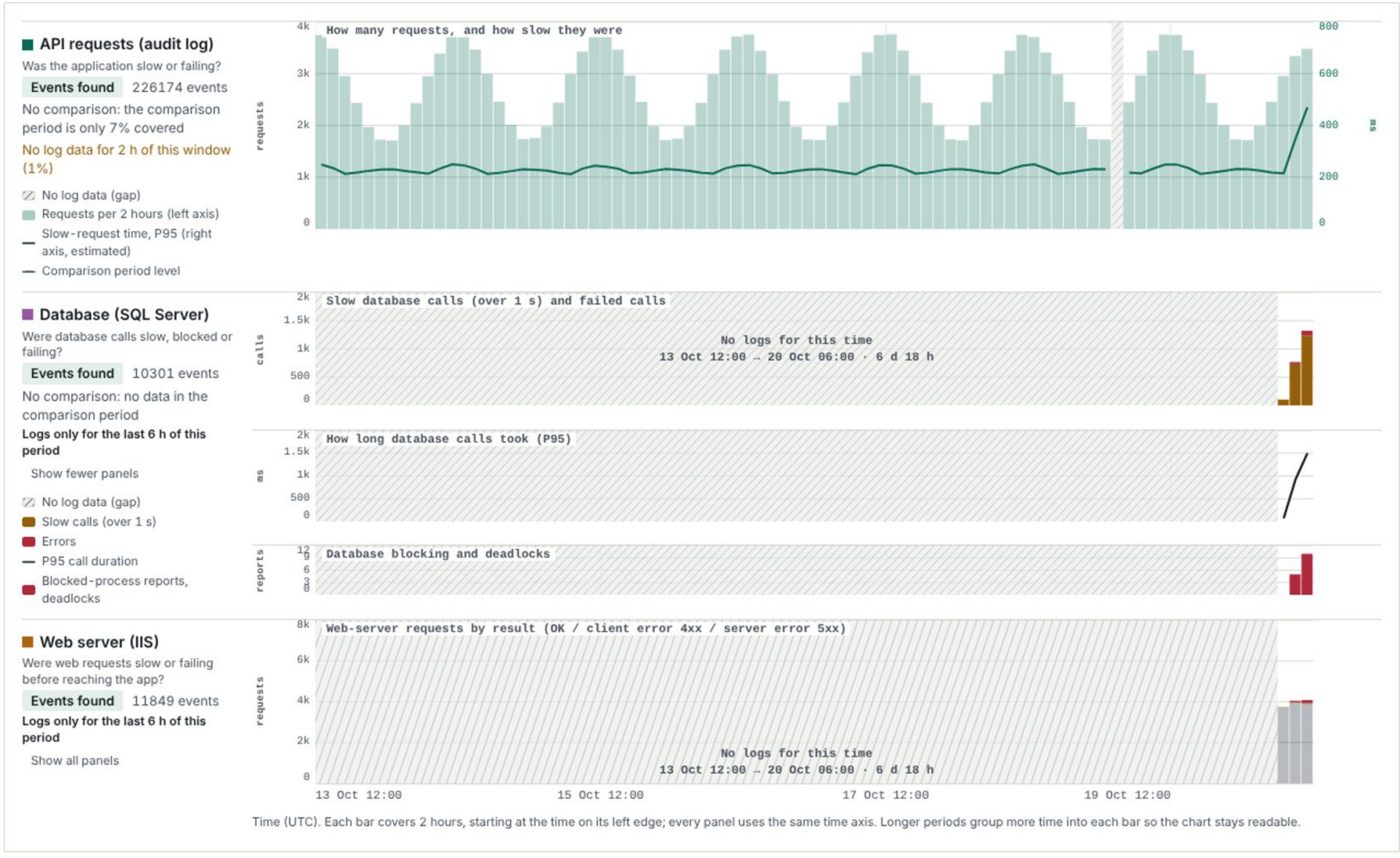
What it shows: Close-up of the seven-day board with the gap labels.
How to reach it: Same screen, chart card.



Time (UTC). Each bar covers 2 hours, starting at the time on its left edge; every panel uses the same time axis. Longer periods group more time into each bar so the chart stays readable.

Image 231-timeline-seven-days-show-all-panels.png

What it shows: A lane with logs for under 15% of the window folds to one short panel; Show all panels expands it.
How to reach it: Press Show all panels on a folded lane.



Time (UTC). Each bar covers 2 hours, starting at the time on its left edge; every panel uses the same time axis. Longer periods group more time into each bar so the chart stays readable.

Image 232-timeline-seven-days-evidence-card.png

What it shows: Evidence card in the seven-day situation.
How to reach it: Same screen, open the evidence card.

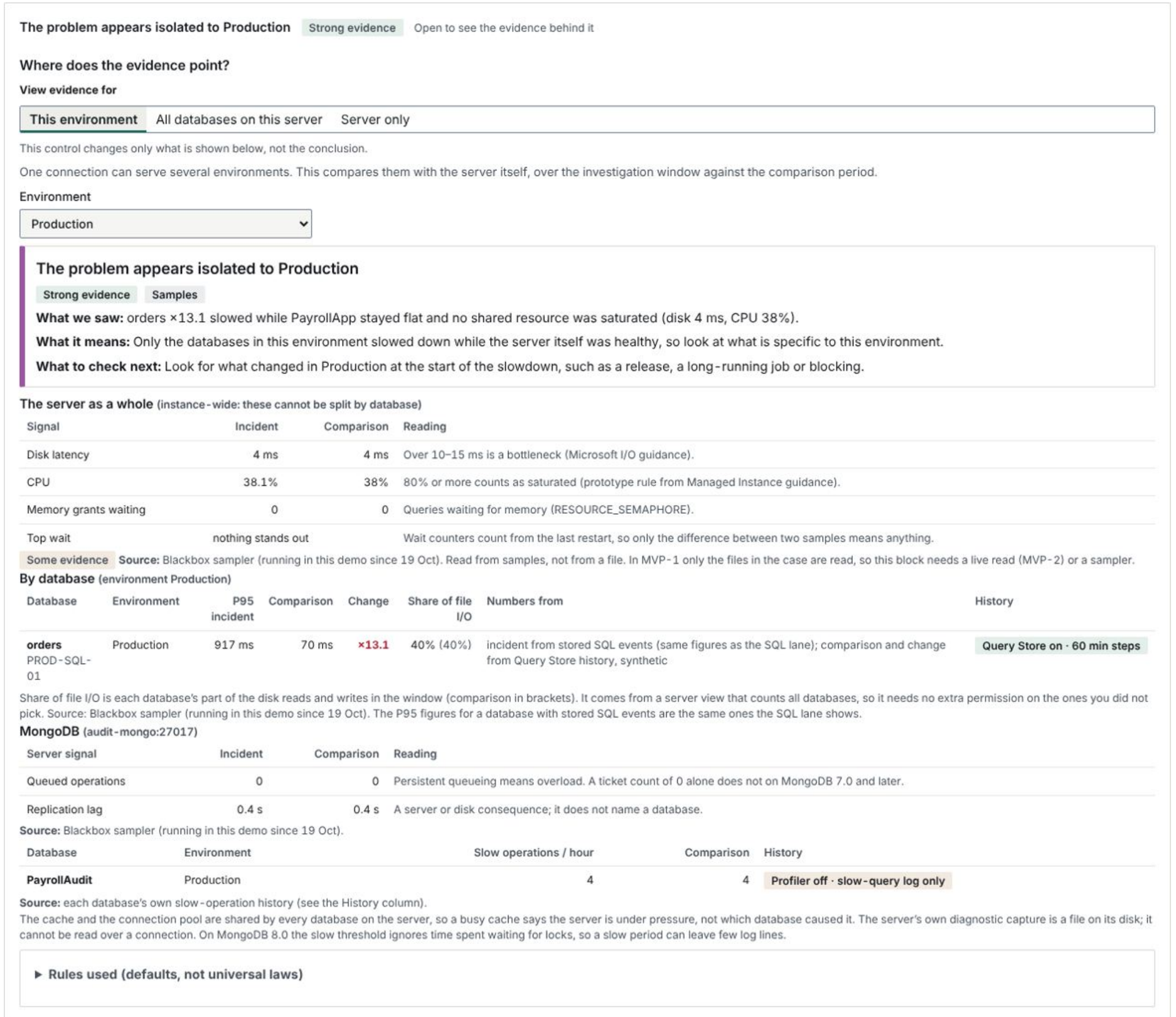


Image 233-timeline-overlap-notice.png

What it shows: The two windows overlap: a non-blocking notice. The timeline still runs.
How to reach it: Scenario w-overlap.

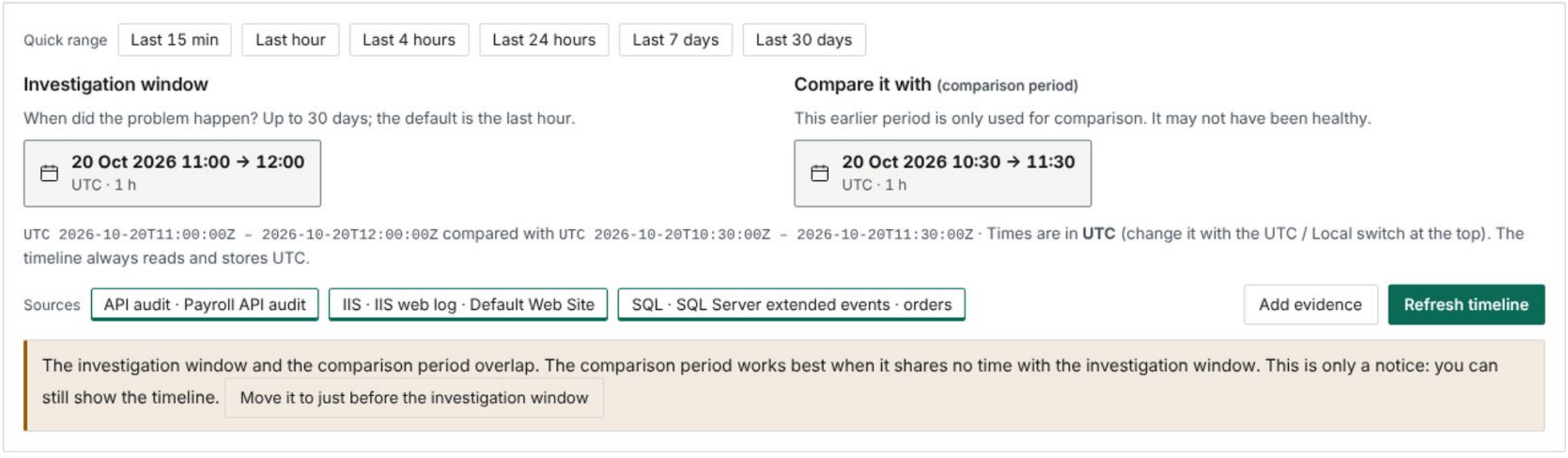


Image 234-timeline-overlap-fix.png

What it shows: Move it to just before the investigation window fixes the overlap in one press.
How to reach it: Press Move it to just before the investigation window.

